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A Mini-Project Report on

“MIDI THEREMIN CONTROLLER”

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Abstract

This project presents the design and development of a MIDI (Musical Instrument Digital Interface) controller using an Arduino Uno microcontroller. MIDI controllers play a crucial role in modern music production by serving as input devices that allow musicians and producers to send digital signals to synthesizers, digital audio workstations (DAWs), and virtual instruments. Unlike traditional musical instruments, a MIDI controller does not produce sound by itself; instead, it transmits digital MIDI messages that control sound-generating hardware or software.

The objective of this project is to create a cost-effective, customizable, and user-friendly MIDI controller tailored for electronic music applications. The system consists of a set of tactile push buttons wired to the Arduino Uno, which processes user input and sends standard MIDI signals through a USB connector. The controller uses the Serial MIDI protocol to interface with external MIDI-compatible devices or a PC. An optional display module can be integrated to provide real-time feedback to the user, replacing conventional LEDs and enhancing the user experience.

This controller is suitable for musicians, hobbyists, and students who seek a simple yet functional tool to interact with music software. It can be expanded with additional features such as rotary knobs, sliders, or touch sensors. Overall, the project demonstrates a practical application of embedded systems in digital music technology, combining electronics, programming, and audio interface knowledge to build a working prototype of a MIDI controller.

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CHAPTER-01

INTRODUCTION

MIDI (Musical Instrument Digital Interface) is a standard communication protocol used in the music industry to connect digital musical instruments, computers, and audio devices. A MIDI controller is an input device that sends MIDI messages to control sound modules, synthesizers, or digital audio workstations (DAWs). Unlike traditional instruments, it does not produce sound on its own but plays a key role in controlling how sound is generated and manipulated in music production.

This project involves the development of a simple and cost-effective MIDI controller using an **Arduino Uno**. The controller uses multiple **push buttons** to send MIDI signals through a **USB connector**, following the standard **Serial MIDI protocol**. Each button represents a specific MIDI note, allowing users to trigger sounds in compatible software or hardware instruments.

The use of Arduino makes the system beginner-friendly and highly customizable. Additionally, an optional **OLED or LCD display** can be included to show feedback, such as button activity or note values. The project demonstrates the integration of electronics, coding, and audio interface design, making it an ideal educational project for understanding embedded systems and digital music technology.

CHAPTER -02

OBJECTIVE

The main objective of this project is to design and develop a simple, low-cost, and functional **MIDI controller** using an **Arduino Uno** microcontroller. The controller will be capable of sending MIDI signals through a **USB connector** to control virtual instruments, synthesizers, or music software on a computer or MIDI-compatible hardware.

The specific objectives of the project are:

1. To understand the **working of MIDI protocol** and its role in digital music production.
2. To design a hardware circuit that includes **buttons** as input devices connected to the Arduino.
3. To program the Arduino to **send MIDI messages** corresponding to button presses using the **Serial MIDI communication protocol**.
4. To provide **real-time visual feedback** to the user using a display screen or LED indicators.
5. To create a modular and expandable MIDI controller that can be improved with features like sliders, knobs, or wireless communication in the future.
6. To apply knowledge of embedded systems, electronics, and programming in a practical music technology application.

CHAPTER-03

METHODOLOGY

➤ BLOCK DIAGARAM

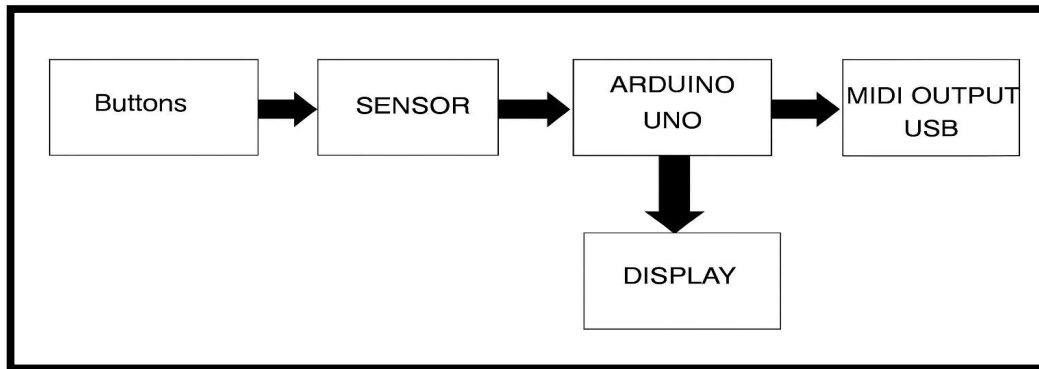


Fig. 3.1 Block Diagram of the project

1. Push Buttons

These are the main input components. Each button is assigned a specific MIDI note. When a button is pressed, it sends a signal to the Arduino Uno indicating that a note should be played or stopped.

2. Arduino Uno

This is the core processing unit of the system. It reads inputs from the buttons, processes the data, and sends appropriate MIDI messages. It acts as a translator between physical input (button press) and digital MIDI signals.

3. USB

This project uses a USB connection to transmit MIDI data directly from the Arduino to the computer. It enables easy integration with digital audio workstations (DAWs). The USB also provides power and data transfer in a single connection.

4. Display (Optional)

An optional display (such as an OLED or LCD) can be added to show which note is being played or give other real-time feedback to the user. This improves usability and visual interaction.

Each component is connected to ensure that when a button is pressed, the system responds by generating a MIDI signal and optionally showing it on the display.

➤ CIRCUIT DIAGRAM

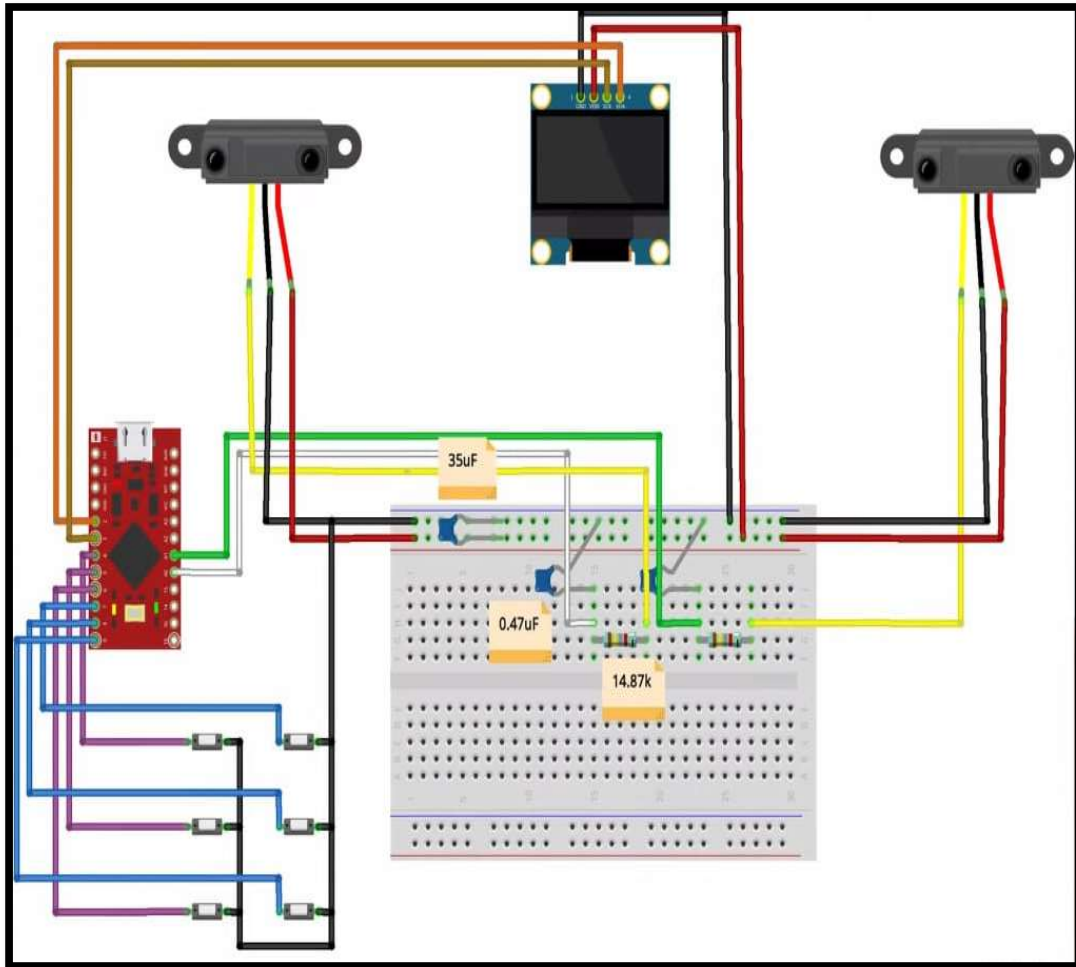


Fig. 3.2 Circuit Diagram

CHAPTER-04

HARDWARE AND SOFTWARE REQUIREMENT

➤ **Hardware Components**

- Arduino Uno
- Push Buttons (x6)
- Ultrasonic Sensors (x2)
- Breadboard
- Jumper Wires
- OLED or LCD Display (Optional)
- USB Cable

➤ **SOFTWARE TOOLS**

- Arduino IDE
- MIDI Library for Arduino
- Serial MIDI Monitor (e.g., Hairless MIDI, MIDI-OX)
- Digital Audio Workstation (e.g., FL Studio, Ableton Live)
- Cakewalk
- MIDI loop

➤ HARDWARE COMPONENTS

Arduino Uno



Fig. 4.1 Arduino Uno

Arduino Uno is a microcontroller board based on the ATmega328P. It is the brain of the MIDI controller, responsible for reading inputs and sending MIDI messages. It has 14 digital I/O pins and supports USB programming.

Features:

- Easy to program using Arduino IDE
- Supports serial communication
- Wide community support and libraries

Ultra Sonic Sensor

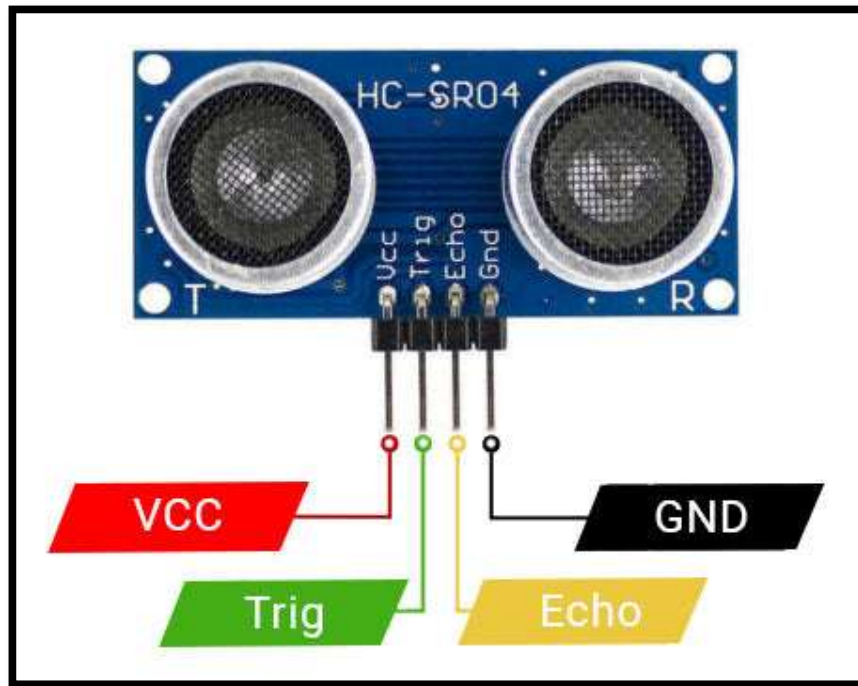


Fig 4.2 Ultra Sonic Sensor

An ultrasonic sensor measures distance using sound waves. In this project, it's used to control MIDI effects based on hand movement or distance. It consists of a transmitter and a receiver.

Features:

- Measures distance from 2 cm to 400 cm
- Non-contact, real-time sensing
- Simple 4-pin connection (VCC, GND, Trigger, Echo)

Bread Board

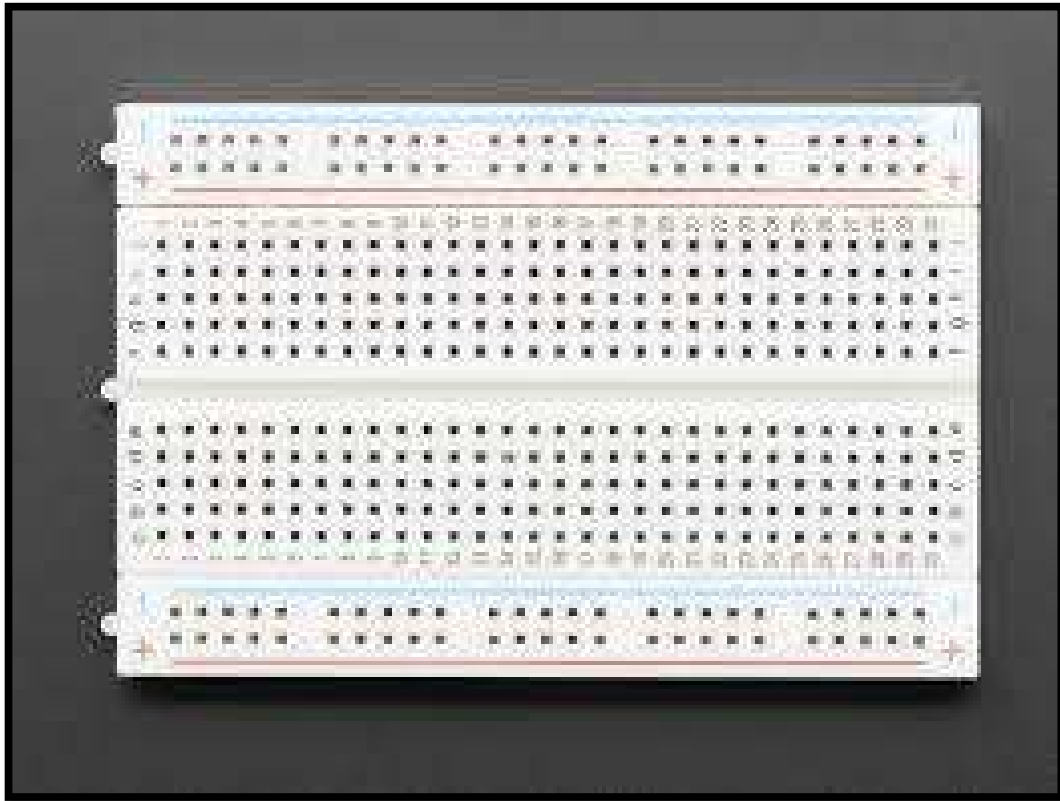


Fig.4.3 Bread Board

A breadboard is a prototyping platform used to build and test electronic circuits without soldering. It allows easy changes in circuit design during development.

Features:

- No soldering required
- Reusable for multiple projects
- Supports both digital and analog components

Jumping Wire

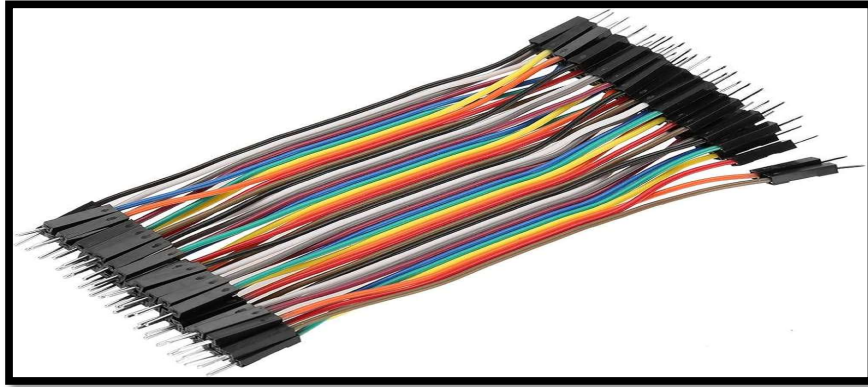


Fig. 4.4 Jumping Wire

Jumper wires are used to connect components on a breadboard or between a breadboard and the Arduino. They come in male-to-male, female-to-male, and female-to-female types.

Features:

- Easy and flexible connections
- Available in various lengths and colours
- Reusable for circuit design

OLED or LCD Display



Fig. 4.5 LCD Display

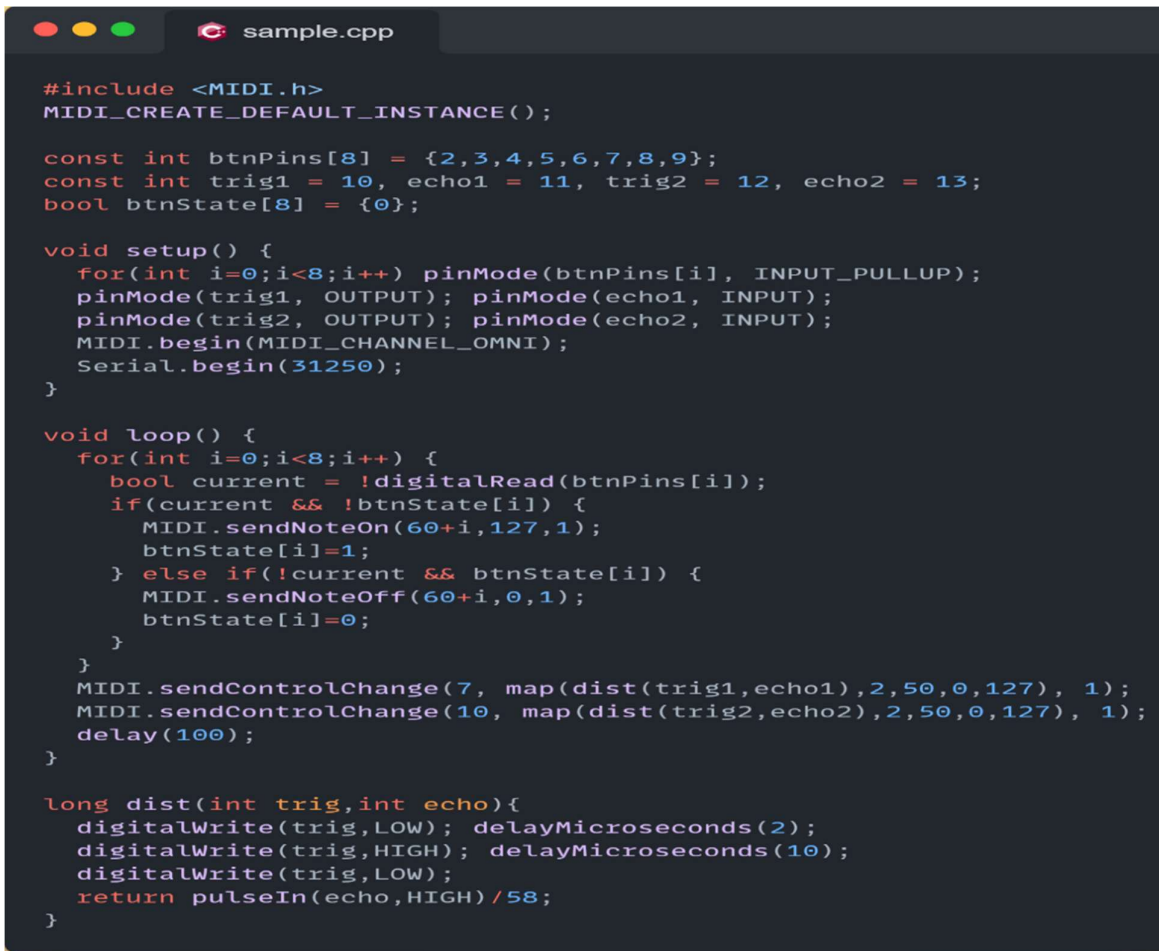
The **16x2 LCD** display is a character-based module that can show two lines with 16 characters each. It is commonly used in embedded projects to display data like sensor values, button activity, or messages. The display connects to Arduino using either parallel pins or an I2C module for simplified wiring.

Features:

- Can display letters, numbers, and symbols
- Easy to interface using Arduino libraries

➤ Software Component

In this MIDI controller project, various software tools played an essential role in development, communication, and sound output. The **Arduino IDE** was used to write, compile, and upload the code to the Arduino Uno board, serving as the main programming platform. The **MIDI Library for Arduino** allowed the board to send and receive standard MIDI messages such as Note On, Note Off, and Control Change. To bridge the serial communication with MIDI, tools like **Hairless MIDI** and **MIDI-OX** were used, converting the Arduino's serial output into MIDI data recognizable by music software. **Digital Audio Workstations** (DAWs) like **FL Studio**, **Ableton Live**, and **Cakewalk** were then used to receive and interpret these MIDI messages, allowing control over virtual instruments and audio effects. Additionally, **loopMIDI** was used to create virtual MIDI ports, enabling seamless software-to-software communication when physical MIDI ports were not available. These tools together provided a complete workflow for testing, debugging, and using the MIDI controller effectively in music production.

A screenshot of the Arduino IDE showing a C++ source code file named 'sample.cpp'. The code is for an Arduino Uno board and uses the MIDI library. It defines 8 buttons (pins 2-9), two triggers (pins 10 and 12), and two echoes (pins 11 and 13). The setup function initializes pins, starts the MIDI library, and sets the serial baud rate to 31250. The loop function reads button states and sends MIDI Note On and Note Off messages. It also reads trigger states and sends MIDI Control Change messages. A 'dist' function is used to calculate the distance between a trigger and an echo, which is then used to generate a pulse width modulation (PWM) signal.

```
#include <MIDI.h>
MIDI_CREATE_DEFAULT_INSTANCE();

const int btnPins[8] = {2,3,4,5,6,7,8,9};
const int trig1 = 10, echo1 = 11, trig2 = 12, echo2 = 13;
bool btnState[8] = {0};

void setup() {
  for(int i=0;i<8;i++) pinMode(btnPins[i], INPUT_PULLUP);
  pinMode(trig1, OUTPUT); pinMode(echo1, INPUT);
  pinMode(trig2, OUTPUT); pinMode(echo2, INPUT);
  MIDI.begin(MIDI_CHANNEL_OMNI);
  Serial.begin(31250);
}

void loop() {
  for(int i=0;i<8;i++) {
    bool current = !digitalRead(btnPins[i]);
    if(current && !btnState[i]) {
      MIDI.sendNoteOn(60+i,127,1);
      btnState[i]=1;
    } else if(!current && btnState[i]) {
      MIDI.sendNoteOff(60+i,0,1);
      btnState[i]=0;
    }
  }
  MIDI.sendControlChange(7, map(dist(trig1,echo1),2,50,0,127), 1);
  MIDI.sendControlChange(10, map(dist(trig2,echo2),2,50,0,127), 1);
  delay(100);
}

long dist(int trig,int echo){
  digitalWrite(trig,LOW); delayMicroseconds(2);
  digitalWrite(trig,HIGH); delayMicroseconds(10);
  digitalWrite(trig,LOW);
  return pulseIn(echo,HIGH)/58;
}
```

Fig.4.6 Sample Source Code of the project

CHAPTER-05

RESULT

SNAPSHOT OF THE PROJECT

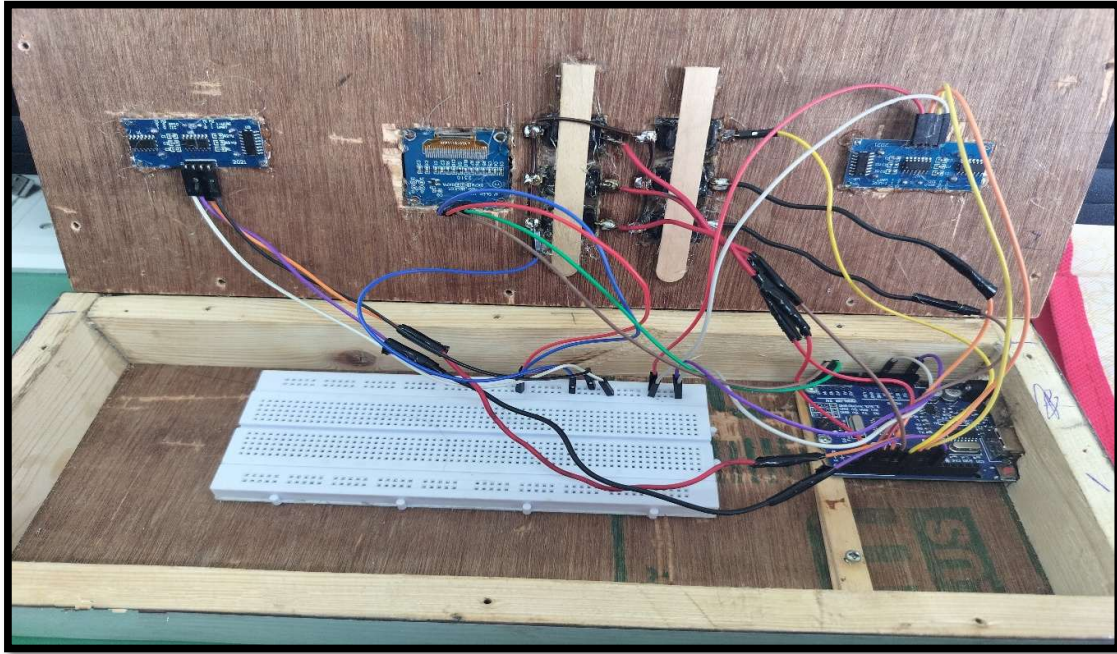


Fig.5.1 Snapshot of circuit arrangement



Fig.5.2 Outlook of the project

CHAPTER-06

ADVANTAGES AND DISADVANTAGES

1 Advantages:

- i. **Low Cost:** Uses affordable components like Arduino Uno and ultrasonic sensors.
- ii. **Customizable:** Easy to modify for different MIDI setups or musical functions.
- iii. **Portable:** Small and lightweight, suitable for live performance setups.
- iv. **Hands-Free Control:** Ultrasonic sensors allow gesture-based control, enhancing creativity.
- v. **Educational:** Great for learning about embedded systems, sensors, and MIDI protocol.
- vi. **Basic Output:** It supports advanced MIDI features like velocity sensitivity or pitch bend.
- vii. **Memory Storage:** Can store and record through the software

2 Disadvantages:

- i. **Limited Precision:** Ultrasonic sensors can give fluctuating readings due to obstacles or noise.
- ii. **Dependency on External Software:** Requires DAWs and MIDI monitors for full functionality.
- iii. **Lag Possibility:** Delays may occur due to serial communication or sensor readings.
- iv. May face compatibility issues with certain DAWs or operating systems without proper drivers.
- v. Limited number of input controls if using basic Arduino boards without external expanders.

Applications

1. **Digital Music Production:**

Used to trigger virtual instruments and sounds in Digital Audio Workstations (DAWs) like FL Studio, Ableton Live, etc.

2. **Live Performances:**

Musicians can control sounds, loops, and effects in real time using buttons or gestures via ultrasonic sensors.

3. **Gesture-Controlled Effects:**

Ultrasonic sensors enable hands-free control over volume, pan, filters, or modulation — ideal for creative expression.

4. **DIY Musical Instruments:**

Forms the foundation of custom instruments designed for experimental or educational purposes.

5. **Music Education Projects:**

Helps students learn about MIDI, sensors, embedded systems, and digital sound synthesis through hands-on building.

FUTURE SCOPE

This project successfully demonstrates the design and implementation of a custom MIDI controller using the Arduino Uno, push buttons, and ultrasonic sensors. The system can send MIDI signals to control virtual instruments and sound effects in real-time, offering both tactile and gesture-based interaction. It provides a cost-effective and flexible solution for musicians, hobbyists, and learners interested in digital music creation. The use of open-source tools and simple components makes it an ideal platform for experimentation and further development.

Additionally, this project serves as a strong foundation for exploring advanced features like wireless MIDI communication or integrating more expressive input methods. With further enhancements, it has the potential to be used in professional music production and live performances. It also encourages innovation in the field of electronic musical instruments.

In the future, this MIDI controller can be upgraded with velocity-sensitive pads, rotary knobs, and faders for more expressive control. Integration of Bluetooth or USB-MIDI can offer wireless capabilities, improving portability and convenience. The addition of preset memory, touch interfaces, or motion sensors could enable dynamic interaction and advanced functionality, expanding its role in both education and real-world music environments.

CONCLUSION

This project successfully demonstrates the design and implementation of a custom MIDI controller using the Arduino Uno, push buttons, and ultrasonic sensors. The system can send MIDI signals to control virtual instruments and sound effects in real-time, offering both tactile and gesture-based interaction. It provides a cost-effective and flexible solution for musicians, hobbyists, and learners interested in digital music creation. The use of open-source tools and simple components makes it an ideal platform for experimentation and further development.

Additionally, this project serves as a strong foundation for exploring advanced features like wireless MIDI communication or integrating more expressive input methods. With further enhancements, it has the potential to be used in professional music production and live performances. It also encourages innovation in the field of electronic musical instruments.

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